

The overall structure of the dashboard

The dashboard is similar to executive dashboard where is to allow non-technical stakeholders to easily understand and get the key insights of the telemarketing campaign's performance. The dashboard will be shared with the sales and marketing team, as well as other interested stakeholders. It provides them with a clear overview of the campaign's results, including statistical models, conversion rates, and time-series forecast to help inform strategic decisions.

The dashboard order follows a story structure from Christopher Booker's plot called "Overing the Monster (Gavin, 2016). The story begins with Exposition (Model Comparison) where the core issue of models' problem introduced. All the statistical models show high accuracy but struggles with low recall. This means that the campaign will miss a lot of potential marketing leads. The next stage is Rising Action (Conversion rate), there is a substantial number of 'unknown' values which limit the model's to fully capture the campaign's customer behaviour. For example, the Education Conversion Rate chart shows that an unknown education level has the highest conversion rate. This indicates that we don't know if high education correlated with higher conversion rates. Removing unknown data will cause data miss leading. The data quality needs improvement from collecting. This will improve model performance.

The next plot is Turning Point (Best Call Days and Job Types for Success) This provides a turning point in the strategy. The sale team can identify optimal days and job types for successful calls, they can target the right audience at the right time. The Climax (Call Duration and Conversion Rate), this dashboard will be discovering how duration and number of contacted are key factors of conversion rate. This becomes the story's secret weapon in overcoming the low success that we mentioned at the start. Finally, Resolution (Forecast

Success Outcome by Month), with insight from the previous dashboard we understand that the company needs to improve the data set, optimal call days and job types and focusing on longer conversations. That will improve the campaign performance.

I have chosen the colour blue-teal because study shows blue colour is found to influence the perception of product financial attractiveness. It is associated with safety, trust, efficiency, duty and logic (international journal of Bank Marketing, 2021) The dashboard uses a darker blue to indicate higher performance values such as higher accuracy or conversion rates. This cue makes it easier for stakeholders to quickly understand and identify the past campaign performance. The light grey background is to keep the dashboard professional and avoid too much colour. Each dashboard has less text, this to minimise distractions, so stakeholders can easily focus on the dashboard. (Zight, 2019)

For the font, Arial is used because it helps to maintain professional appearance. It is well known for its clean lines and easy-to-read characters. (Bogdan Radusinovic, 2024) The summary text box was included in to provide actionable insights directly from the dashboard. This then translates from the visual data into easy and understandable takeaway points. This then ensure that both technical and non-technical stakeholders can understand and make precise data-driven decisions.

The use of filters helps users to explore data themselves and make faster decisions and increase user engagement. The hand on experience promotes a deeper understanding of the data and helps users feel more connected to the information. (Chio,2022)

Details of the choices made:

Model Comparing				
Classification tree	Accuracy 92.25%	Precision 74.09%	Recall 43.47%	F-score 52.11%
Linear Regression	Accuracy 91.53%	Precision 70.29%	Recall 37.39%	F-score 61.11%
Logistic Regression	Accuracy 91.70%	Precision 72.89%	Recall 36.78%	F-score 65.86%

Figure 1

Figure 1 provides a side-by-side comparison of three models (Classification Tree, Linear Regression and Logistic Regression) This is to compare the performance metrics such as accuracy, precision, recall and F-score. The blue-teal gradient highlights the models from the highest accuracy to the lowest accuracy. This makes it easier for non-technical users to quickly interpret which model performs the best overall.

		Prediction	
		No	Yes
Actual	No	True Negative 2,666	False Negative 206
	Yes	False Positive 52	True Positive 123

Figure 2

The confusion matrix hasn't changed the form to different chart. This standardised matrix form ensures that the data presentation remains clear and intuitive. The data has filtered by selecting the model from Figure 1. Users can easily compare blue for correct prediction and light red for incorrect prediction. This approach helps stakeholders identify areas for improvement, such as increasing the number of true positives and minimising errors in other sections. It provides insights into the model's performance issues, which is crucial for the marketing team as they seek to determine the best model for predicting successful outcomes in the campaign.

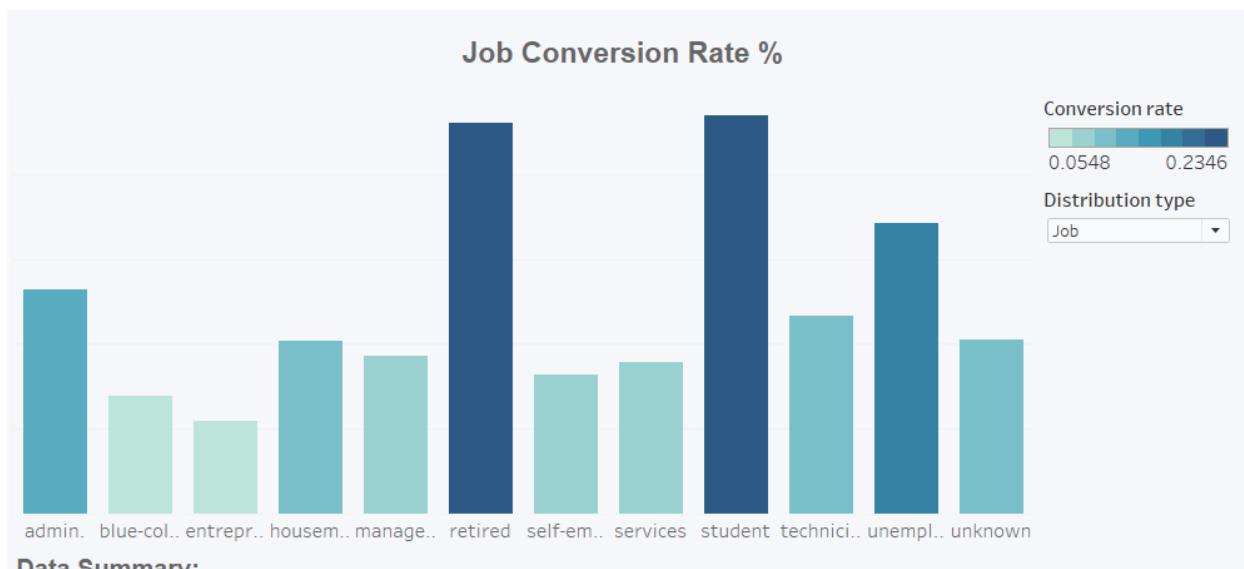


Figure 3

The use of single colour shade consistently is to avoid the rainbow effect (Nebula, 2013), which matches the rest of the dashboard's theme. Data labels are hidden from the individual bars to keep the dashboard minimal, but the conversion rate range is clearly marked in the colour legend on the right of Figure 3. This interactive filter enables users to easily compare how different factors impact conversion rate. Additionally, it maintains the visual cleanliness and a minimalistic design. This design choice helps the marketing team identify new opportunities and

guides the data collection team in refining their data-gathering processes to improve data quality and insights.

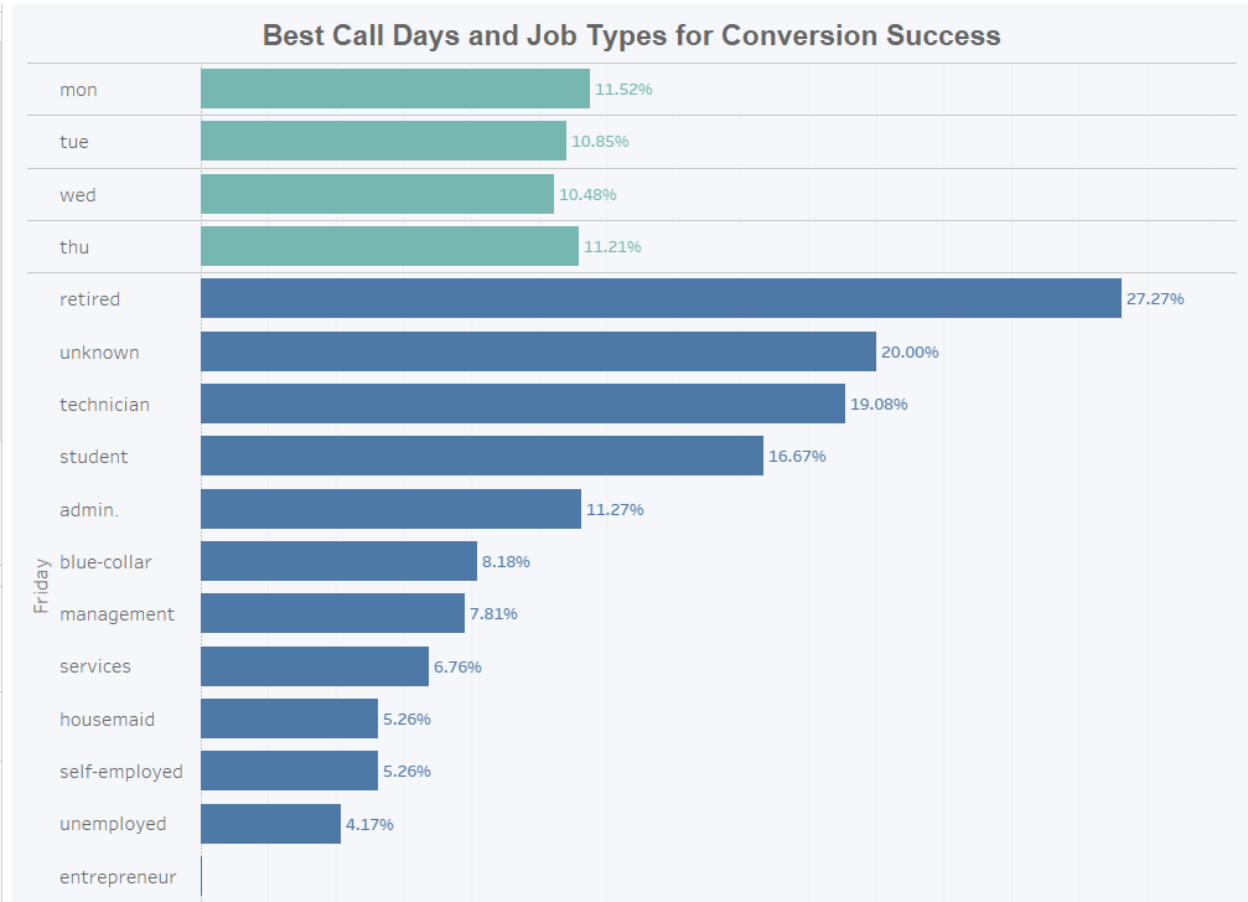


Figure 4

Figure 4 is a dashboard that allows users to drill down into more detailed information by clicking on a specific day or job type. This dashboard is for the sales team to prioritise customer outreach. The length of the bars makes it easy to compare conversion rates across different categories, while the values are sorted in descending order, enabling users to quickly identify which job types have the highest conversion rates, such as retired individuals. This design encourages user interaction with the dashboard, making the experience more engaging and insightful. The interactive features keep the user interested by allowing them to explore and understand the data dynamically.

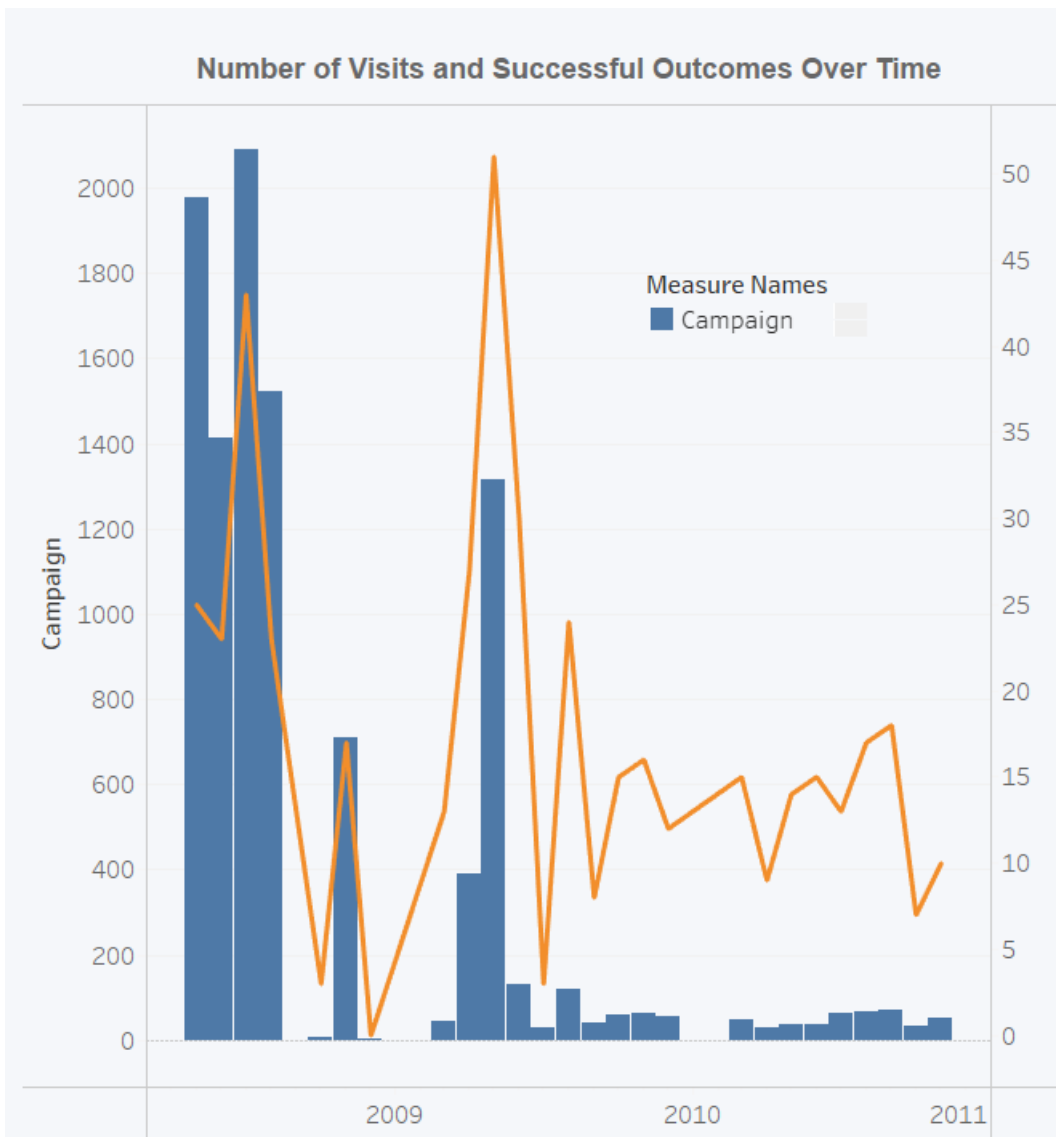


Figure 5 is a combo chart (Elena, 2024) that combines a bar chart and a line chart to compare the relationship between the number of contact visits and successful outcomes over time. The line chart representing outcomes is coloured orange to make it stand out. This allows stakeholders to easily observe the trend line. The bar chart is coloured blue to maintain consistency with the overall dashboard colour scheme. This combination shows how changes in the number of visits correlate with successful outcomes over timeline 2009 – 2011.

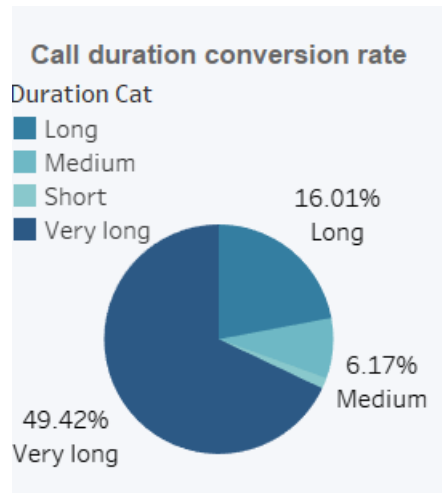


Figure 6, the pie chart, illustrates the proportion of conversion rates across various call duration ranges. The layout makes it easy to identify that the "very long" duration for nearly three-quarters of the total conversions. With the dark blue indicating the highest conversion rate consistency, helps stakeholders quickly interpret the data from the area of the pie.

Figure 6

Figure 5 and 6 are important to sale team to see how important the engaging call and contacting customers relate to outcome. This will encourage sales team to have more meaningful call.

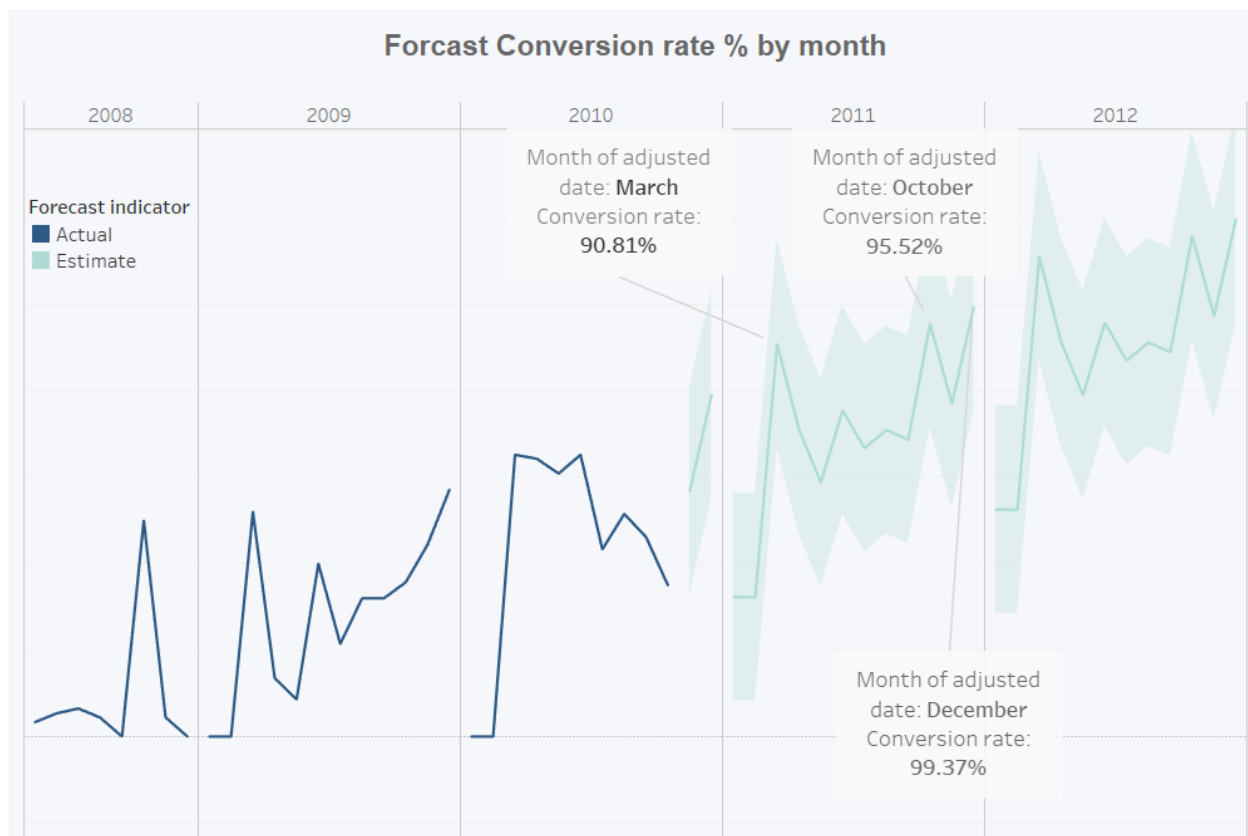


Figure 7

Figure 7 is a line chart displaying forecast conversion rate trends over time with dark blue representing actual conversion rate and green for estimated rate to help stakeholders visually distinguish between real performance and projections. Annotations emphasise key points. This allows users to interpret trends easily. This dashboard is crucial for the marketing team as it provides insights into the best time to launch future campaigns.

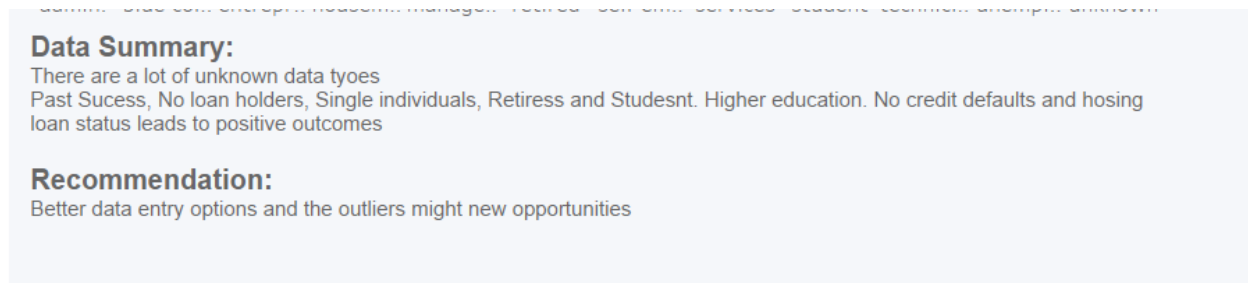


Figure 8

Finally, static data and annotations are incorporated into each dashboard (e.g. Figure 8) to help stakeholders quickly grasp the key points without being overwhelmed by excessive information. This approach ensures that the main insights are clear and accessible, allowing stakeholders to understand the critical findings briefly without needing to reinterpret the data.

References

Bogdan Radusinovic (2014) The Best Fonts For Tables That You Should Be Using. Available from: <https://wpdatatables.com/best-fonts-for-tables/#:~:text=Arial%2C%20Helvetica%2C%20and%20Verdana%20are,or%20designing%20a%20web%20dashboard>. [Accessed 15 October 2024]

Chip (2022) What are the benefits of an interactive over a static one? Available from: <https://www.quanthub.com/what-are-the-benefits-of-an-interactive-dashboard-over-a-static-one/>. [Accessed 18 October 2024]

Elena Kosourava (2024) How to Create and Format a Combo Chart in Excel. Available from: <https://www.datacamp.com/tutorial/mastering-combo-charts-in-excel> [Accessed 18 October 2024]

Gavin McMahon (2016) Story structure – the hidden framework that hangs your story together. Available from: <https://www.presentation-guru.com/on-structure-the-hidden-framework-that-hangs-your-story-together/> [Accessed 13 October 2024]

International Journal of Bank Marketing (2021) Blue and red in financial documents: the influence on attentional mechanisms and behavior. Available from: <https://www.emerald.com/insight/content/doi/10.1108/IJBM-10-2020-0527/full/html> .[Accessed 29 September 2024]

Olivvier Cathrin.(2020) Decision trees, flow diagrams, sankeys in Tableau... here is a solution !!!..Available from: <https://community.tableau.com/s/question/0D54T00000C5Q1ISAV/decision-trees-flow-diagrams-sankeys-in-tableau-here-is-a-solution-> .[Accessed 29 September 2024]

OneNumber (2023) Apply Filter to Multiple Worksheets in Tableau (Quick Tip). Available from: <https://www.youtube.com/watch?v=mq32Ln-2cNg&t=49s> [Accessed 08 October 2024]

Technology mart (2016) Trend Lines in Tableau. Available from:

<https://www.youtube.com/watch?v=FBK4aT2D45Y&t=39s> [Accessed 07 October 2024]

Zight (2019) Form Design UX: The 7 Best Practice for User Friendly Forms. Available form:

<https://zight.com/blog/form-design-ux/> [Accessed 11 October 2024]